

Journal Pre-proof

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PII: S2405-9595(26)00143-8
DOI: <https://doi.org/10.1016/j.ict.2026.08.001>
Reference: ICTE 1092

To appear in: *ICT Express*

Received date: 2 December 2025

Revised date: 15 June 2026

Accepted date: 3 August 2026

Please cite this article as: X. Ni, C. Hu, X. Yu et al., Energy-efficient relay-assisted THz V2X networks with integrated sensing and communication, *ICT Express* (2026), doi: <https://doi.org/10.1016/j.ict.2026.08.001>.

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Energy-Efficient Relay-Assisted THz V2X Networks with Integrated Sensing and Communication

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Abstract

Terahertz (THz) V2X networks with integrated sensing and communication (ISAC) face severe blockage and energy constraints in urban environments. This letter studies relay-assisted communication with vehicle-side sensing QoS constraints. We formulate a joint beamforming optimization problem to minimize power consumption under strict communication and sensing constraints. An efficient iterative algorithm utilizing Matrix-Extended Generalized Lagrangian Dual Transformation (MEGLDT) and Block Coordinate Descent (BCD) is developed. Simulations demonstrate that the proposed scheme yields significant power savings compared to benchmarks, particularly in high-density scenarios. These findings validate the framework's scalability and efficacy for next-generation V2X applications.

Keywords: Integrated sensing and communication, Terahertz, Vehicle-to-Everything, Power minimization.

1. Introduction

With the rapid advancement of the Internet of Vehicles (IoV) and Vehicle-to-Everything (V2X), emerging B5G and 6G networks are increasingly tasked with supporting the rapid processing and transmission of massive data volumes [1]. In urban environments, high-rise buildings and other obstacles often disrupt direct communication links between vehicles and base stations (BSs), a challenge that becomes particularly pronounced when leveraging the terahertz (THz) frequency band. THz communication offers high data rates for V2X applications [2]; however, its high path loss and sensitivity to blockages severely limit its effectiveness over long distances. Reconfigurable intelligent surfaces (RISs) are promising for blockage mitigation, but their practical use in high-mobility V2X still faces deployment, control, and CSI-acquisition challenges; moreover, relay/repeater-type active forwarding has higher deployment maturity and practical implementation feasibility [3]. To address this, relay-assisted communication has emerged as a compelling solution, where intermediate relay nodes mitigate signal blockages and en-

hance reliability by providing alternative transmission paths [4].

In V2X networks, vehicles must simultaneously perform communication and sensing tasks to support autonomous driving and intelligent traffic management [5]. These dual requirements necessitate a system capable of supporting both integrated sensing and communication (ISAC) functionalities while optimizing resource utilization [6]. Power efficiency is especially vital in IoV, as it directly impacts vehicle battery life and operational costs [7]. Existing studies have explored THz communication in vehicular networks [8], hybrid beamforming for mmWave ISAC [9], RIS-aided mmWave ISAC [10], and ISAC techniques for spectrum and hardware efficiency [11, 12]. Different from these works, this letter focuses on energy-efficient beamforming in relay-assisted THz V2X networks under communication-rate and vehicle-side sensing-QoS constraints.

In this letter, we propose a relay-assisted THz V2X network with integrated communication and vehicle-side sensing. Our primary goal is to minimize the total power consumption while meeting quality-of-service (QoS) constraints for both communication and sensing performance. To this end, we design and optimize the transmit beamforming matrices for vehicles as well as

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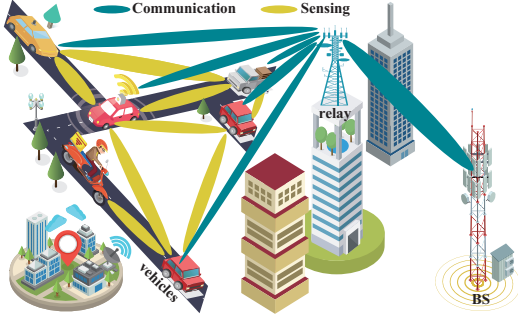


Figure 1. Application scenario of the relay-assisted THz V2X network with integrated sensing and communication.

the cooperative relay, ensuring efficient resource utilization. As a result, we formulate a power minimization problem and propose an efficient algorithm that integrates the Matrix-Extended Generalized Lagrangian Dual Transformation (MEGLDT) with the block coordinate descent (BCD) technique to deal with it. The main contributions of this letter are as follows:

- A relay-assisted THz V2X architecture is proposed to improve V2X communication reliability while satisfying vehicle-side sensing QoS constraints.
- We formulate a joint beamforming optimization problem to minimize total power consumption, subject to strict constraints on communication rate and sensing performance.
- An efficient iterative algorithm based on the MEGLDT and BCD is developed to solve the resulting non-convex problem.
- Simulation results demonstrate that the proposed scheme achieves significant power savings compared to benchmark techniques, particularly in high-density vehicular scenarios.

2. System Model and Problem Formulation

We consider a multi-vehicle THz V2X network where a multi-antenna relay assists the V2R/R2B transmission of the superimposed sensing-and-communication waveform, while the sensing QoS is evaluated at the vehicle side. As depicted in Fig. 1, the system comprises K vehicles, one relay, and one BS. Each vehicle has M antennas, the relay has N_r antennas, and the BS has N_b antennas. Due to long distances or obstacles, direct vehicle-to-BS (V2B) links are unavailable. Let $\mathbf{H}_k \in \mathbb{C}^{N_r \times M}$ denote the vehicle-to-relay channel for vehicle k , $k \in \mathcal{K} \triangleq \{1, \dots, K\}$, and let $\mathbf{G} \in \mathbb{C}^{N_b \times N_r}$ denote the relay-to-BS channel.

2.1. THz Channel Model

We adopt the THz channel model developed by [13], which leverages a geometric framework characterized by sparsity and a limited number of dominant paths. The V2R channel is modeled as

$$\mathbf{H}_k = \sum_{n=1}^{N_{\text{sub}}} \sum_{l=1}^L \alpha_{k,r,l}^n \mathbf{a}_r(\theta_r^l, \phi_r^l) \mathbf{a}_k^H(\theta_t^l, \phi_t^l), k \in \mathcal{K}, \quad (1)$$

where N_{sub} is the number of subbands, L is the number of dominant paths, and $\alpha_{k,r,l}^n$ is the complex path gain of the l -th path in the n -th subband for vehicle k . For uniform linear arrays, the array responses can be written as

$$\mathbf{a}_k(\theta_t^l) = \frac{1}{\sqrt{M}} \left[1, e^{j\frac{2\pi}{\lambda} d \sin \theta_t^l}, \dots, e^{j(M-1)\frac{2\pi}{\lambda} d \sin \theta_t^l} \right]^T, \quad (2)$$

$$\mathbf{a}_r(\theta_r^l) = \frac{1}{\sqrt{N_r}} \left[1, e^{j\frac{2\pi}{\lambda} d \sin \theta_r^l}, \dots, e^{j(N_r-1)\frac{2\pi}{\lambda} d \sin \theta_r^l} \right]^T, \quad (3)$$

where $\lambda = c/f_c$ is the wavelength and d is the antenna spacing. Similarly, the R2B channel $\mathbf{G}$ is modeled by

$$\mathbf{G} = \sum_{n=1}^{N_{\text{sub}}} \sum_{l=1}^L \alpha_{g,l}^n \mathbf{a}_b(\theta_r^l, \phi_r^l) \mathbf{a}_r^H(\theta_t^l, \phi_t^l), \quad (4)$$

where $\alpha_{g,l}^n$ is the complex path gain of the l -th path in the n -th subband.

2.2. Relay-Assisted Communication Model

In the first phase, all vehicles simultaneously transmit their sensing and communication signals to the relay. The transmitted signal from vehicle k is expressed as

$$\mathbf{x}_k = \mathbf{W}_{k,s} \mathbf{s}_k + \mathbf{W}_{k,c} \mathbf{u}_k, \quad (5)$$

where $\mathbf{W}_{k,s} \in \mathbb{C}^{M \times I}$ and $\mathbf{W}_{k,c} \in \mathbb{C}^{M \times J}$ are the beamforming matrices for sensing and communication, respectively. $\mathbf{s}_k \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_I)$ and $\mathbf{u}_k \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_J)$ denote the sensing and communication symbols, respectively. Thus, the transmit power of vehicle k is

$$P_k = \|\mathbf{W}_{k,s}\|_F^2 + \|\mathbf{W}_{k,c}\|_F^2, k \in \mathcal{K}. \quad (6)$$

The received signal at the relay is given by

$$\mathbf{y}_r = \sum_{k=1}^K \mathbf{H}_k \mathbf{x}_k + \mathbf{n}_r, \quad (7)$$

where $\mathbf{n}_r \sim \mathcal{CN}(\mathbf{0}, \sigma_r^2 \mathbf{I}_{N_r})$ denotes the additive white Gaussian noise (AWGN) at the relay.

In the second phase, the relay amplifies and forwards the received signal to the BS using the beamforming matrix $\mathbf{W}_r \in \mathbb{C}^{N_r \times N_r}$, and its corresponding transmit signal is

$$\mathbf{x}_r = \mathbf{W}_r \mathbf{y}_r = \sum_{k=1}^K \mathbf{W}_r \mathbf{H}_k \mathbf{x}_k + \mathbf{W}_r \mathbf{n}_r. \quad (8)$$

Based on (8), the relay's transmit power is

$$P_r = \sum_{k=1}^K \left(\|\mathbf{W}_r \mathbf{H}_k \mathbf{W}_{k,s}\|_F^2 + \|\mathbf{W}_r \mathbf{H}_k \mathbf{W}_{k,c}\|_F^2 \right) + \sigma_r^2 \|\mathbf{W}_r\|_F^2. \quad (9)$$

From (8), the received signal at the BS is expressed as

$$\mathbf{y}_b = \mathbf{G} \mathbf{x}_r + \mathbf{n}_b = \sum_{k=1}^K \mathbf{G} \mathbf{W}_r \mathbf{H}_k \mathbf{W}_{k,c} \mathbf{u}_k + \sum_{k=1}^K \mathbf{G} \mathbf{W}_r \mathbf{H}_k \mathbf{W}_{k,s} \mathbf{s}_k + \mathbf{G} \mathbf{W}_r \mathbf{n}_r + \mathbf{n}_b, \quad (10)$$

where $\mathbf{n}_b \sim \mathcal{CN}(0, \sigma_b^2 \mathbf{I}_{N_b})$ is the AWGN at the BS. Note that $\mathbf{y}_b$ encompasses both the communication and sensing components, along with the amplified noise introduced at the relay. Since the sensing waveforms are known at the BS, an ideal receiver-side cancellation assumption is adopted for tractable communication decoding. The problem remains coupled because the sensing and communication beamformers share the vehicle power budgets, both components contribute to the relay transmit power, and the relay beamformer is jointly optimized with the vehicle beamformers. Therefore, the effective signal for communication is

$$\mathbf{y}'_b = \sum_{k=1}^K \mathbf{G} \mathbf{W}_r \mathbf{H}_k \mathbf{W}_{k,c} \mathbf{u}_k + \mathbf{G} \mathbf{W}_r \mathbf{n}_r + \mathbf{n}_b. \quad (11)$$

According to (11), the achievable communication rate for vehicle k is then given by

$$R_k = \log_2 \det \left(\mathbf{I}_J + \mathbf{W}_{k,c}^H \mathbf{H}_k^H \mathbf{W}_r^H \mathbf{G}^H \mathbf{N}_k^{-1} \mathbf{G} \mathbf{W}_r \mathbf{H}_k \mathbf{W}_{k,c} \right), \quad (12)$$

where

$$\mathbf{N}_k = \sum_{j \neq k} \mathbf{G} \mathbf{W}_r \mathbf{H}_j \mathbf{W}_{j,c} \mathbf{W}_{j,c}^H \mathbf{H}_j^H \mathbf{W}_r^H \mathbf{G}^H + \sigma_r^2 \mathbf{G} \mathbf{W}_r \mathbf{W}_r^H \mathbf{G}^H + \sigma_b^2 \mathbf{I}_{N_b}. \quad (13)$$

2.3. Vehicle-Side Sensing Model

The sensing task focuses on a single target per vehicle, leveraging both sensing and communication signals. The communication signal does not interfere with sensing at the vehicle receiver due to prior knowledge or separation techniques [14]. For the sensing task, each vehicle estimates environmental parameters using the echo signal [13]

$$\mathbf{r}_k = \beta_k \mathbf{\Lambda}_k \mathbf{x}_k + \mathbf{v}_k = \beta_k \mathbf{\Lambda}_k (\mathbf{W}_{k,s} \mathbf{s}_k + \mathbf{W}_{k,c} \mathbf{u}_k) + \mathbf{v}_k, \quad (14)$$

where β_k is the reflection coefficient, $\mathbf{\Lambda}_k$ is the array response matrix, and $\mathbf{v}_k \sim \mathcal{CN}(\mathbf{0}, \sigma_v^2 \mathbf{I}_M)$ is AWGN at vehicle k .

Following information-theoretic ISAC metrics, we use the sensing rate (SR) to quantify the target-related environmental information extracted from the echo observation, while the communication rate (CR) describes

user data delivery. This SR-CR description is consistent with mutual-information-based ISAC formulations and provides a common rate-domain QoS measure for joint beamforming design [15]. The SR also increases with the effective sensing SINR, a key indicator for detection and localization. Thus, the sensing signal estimation rate at vehicle k is given by [16]

$$\omega_k = \log_2 \det \left(\mathbf{I}_{I+J} + \frac{|\beta_k|^2}{\sigma_v^2} \mathbf{\Sigma}_k^H \mathbf{\Lambda}_k^H \mathbf{\Lambda}_k \mathbf{\Sigma}_k \right) \quad (15)$$

where $\mathbf{\Sigma}_k = [\mathbf{W}_{k,s}, \mathbf{W}_{k,c}]$.

2.4. Problem Formulation

In this work, we aim to minimize the total power consumption while satisfying the constraints on communication rates and sensing performance. Consequently, the optimization problem is formulated as:

$$\min_{\{\mathbf{W}_{k,s}, \mathbf{W}_{k,c}\}, \mathbf{W}_r} \sum_{k=1}^K P_k + P_r \quad (16a)$$

$$\text{s.t. } R_k \geq R_{\min}, \quad \forall k \in \mathcal{K}, \quad (16b)$$

$$\omega_k \geq \omega_k^{\min}, \quad \forall k \in \mathcal{K}, \quad (16c)$$

$$P_k \leq P_k^{\max}, \quad \forall k \in \mathcal{K}, \quad (16d)$$

$$P_r \leq P_r^{\max}, \quad (16e)$$

where $P_k^{\max}$ and $P_r^{\max}$ are the maximum transmit powers of vehicle k and the relay, respectively. Constraints (16b) and (16c) ensure the communication rate and sensing performance requirements, while (16d) and (16e) are the individual power constraints for vehicles and the relay.

3. Proposed Iterative Algorithm Based on MEGLDT and BCD

The formulated optimization problem (16) is highly nonconvex due to coupled variables within log-determinant functions. To address it, we propose an iterative algorithm leveraging the MEGLDT combined with the BCD method. This framework effectively disentangles the coupled variables, transforming non-convex constraints into a sequence of tractable convex formulations.

According to the MEGLDT principle [17], the general log-determinant function of the form $f(\mathbf{X}) = \ln \det(\mathbf{I} + \mathbf{A}(\mathbf{X})^H \mathbf{B}^{-1}(\mathbf{X}) \mathbf{A}(\mathbf{X}))$ can be lowerbounded by constructing a suitable surrogate function through the introduction of auxiliary variables. Since these auxiliary variables admit closedform solutions, MEGLDT avoids the high computational complexity that would arise from directly optimizing the logdeterminant expression. Consequently, MEGLDT is able to transform

the original nonconvex problem into its blockconvex form and solve it with low complexity.

We first apply this transformation to the communication rate constraint in (16b). Let $\mathbf{H}_{k,\text{eff}} = \mathbf{G}\mathbf{W}_r\mathbf{H}_k$ denote the equivalent cascaded channel from vehicle k to the BS. The rate for vehicle k is given by:

$$R_k = \log_2 \det(\mathbf{I}_J + \mathbf{W}_{k,c}^H \mathbf{H}_{k,\text{eff}}^H \mathbf{N}_k^{-1} \mathbf{H}_{k,\text{eff}} \mathbf{W}_{k,c}). \quad (17)$$

Mapping this to the MEGLDT form, we identify $\mathbf{A}(\mathbf{X}) \rightarrow \mathbf{H}_{k,\text{eff}} \mathbf{W}_{k,c}$ and $\mathbf{B}(\mathbf{X}) \rightarrow \mathbf{N}_k$. By introducing auxiliary variables $\mathbf{E}_k \in \mathbb{C}^{J \times J}$ and $\mathbf{Z}_k \in \mathbb{C}^{N_b \times J}$, we derive the surrogate function (scaled by $\ln 2$):

$$f_k^{\text{surr}} = \ln \det(\mathbf{I}_J + \mathbf{E}_k) - \text{tr}(\mathbf{E}_k) + 2\Re \left\{ \text{tr} \left(\mathbf{Z}_k^H \mathbf{H}_{k,\text{eff}} \mathbf{W}_{k,c} \right) - \text{tr} \left(\mathbf{Z}_k^H \mathbf{R}_{\text{total}} \mathbf{Z}_k (\mathbf{I}_J + \mathbf{E}_k)^{-1} \right) \right\}, \quad (18)$$

where $\mathbf{R}_{\text{total}} = \sum_{j=1}^K \mathbf{H}_{j,\text{eff}} \mathbf{W}_{j,c} \mathbf{W}_{j,c}^H \mathbf{H}_{j,\text{eff}}^H + \sigma_r^2 \mathbf{G}\mathbf{W}_r \mathbf{W}_r^H \mathbf{G}^H + \sigma_b^2 \mathbf{I}_{N_b}$ represents the total received covariance matrix. For fixed beamformers, the optimal auxiliary updates are:

$$\mathbf{E}_k^* = \mathbf{W}_{k,c}^H \mathbf{H}_{k,\text{eff}}^H \mathbf{N}_k^{-1} \mathbf{H}_{k,\text{eff}} \mathbf{W}_{k,c}, \quad (19a)$$

$$\mathbf{Z}_k^* = \mathbf{R}_{\text{total}}^{-1} \mathbf{H}_{k,\text{eff}} \mathbf{W}_{k,c} (\mathbf{I}_J + \mathbf{E}_k^*)^{-1}. \quad (19b)$$

Critically, when $\mathbf{E}_k$ and $\mathbf{Z}_k$ are fixed, the negative trace term in (18) is a concave function of $\mathbf{W}_{k,c}$, rendering the constraint $f_k^{\text{surr}} \geq R_{\min} \ln 2$ convex.

Subsequently, we focus on the processing of nonconvex constraint (16c). Mapping the sensing rate equation to the MEGLDT form with $\mathbf{A} \rightarrow \frac{\beta_k}{\sigma_v} \mathbf{\Lambda}_k \mathbf{\Sigma}_k$ and $\mathbf{B} \rightarrow \mathbf{I}_M$, we introduce $\mathbf{F}_k \in \mathbb{C}^{(I+J) \times (I+J)}$ and $\mathbf{U}_k \in \mathbb{C}^{M \times (I+J)}$. The surrogate sensing function is:

$$g_k^{\text{surr}} = \ln \det(\mathbf{I}_{I+J} + \mathbf{F}_k) - \text{tr}(\mathbf{F}_k) + 2\Re \left\{ \text{tr} \left(\mathbf{U}_k^H \tilde{\mathbf{H}}_k \mathbf{\Sigma}_k \right) - \text{tr} \left(\mathbf{U}_k^H \left(\tilde{\mathbf{H}}_k \mathbf{\Sigma}_k \mathbf{\Sigma}_k^H \tilde{\mathbf{H}}_k^H + \sigma_v^2 \mathbf{I}_M \right) \mathbf{U}_k (\mathbf{I}_{I+J} + \mathbf{F}_k)^{-1} \right) \right\}, \quad (20)$$

where $\tilde{\mathbf{H}}_k = \beta_k \mathbf{\Lambda}_k$. The optimal updates are:

$$\mathbf{F}_k^* = \sigma_v^{-2} \mathbf{\Sigma}_k^H \tilde{\mathbf{H}}_k^H \tilde{\mathbf{H}}_k \mathbf{\Sigma}_k, \quad (21a)$$

$$\mathbf{U}_k^* = \left(\tilde{\mathbf{H}}_k \mathbf{\Sigma}_k \mathbf{\Sigma}_k^H \tilde{\mathbf{H}}_k^H + \sigma_v^2 \mathbf{I}_M \right)^{-1} \tilde{\mathbf{H}}_k \mathbf{\Sigma}_k (\mathbf{I}_{I+J} + \mathbf{F}_k^*)^{-1}. \quad (21b)$$

Ultimately, we obtain the following convex reformulation of problem (16):

$$\begin{aligned} \min_{\{\mathbf{W}_{k,s}, \mathbf{W}_{k,c}\}, \mathbf{W}_r} \quad & \sum_{k=1}^K P_k + P_r \\ \text{s.t.} \quad & f_k^{\text{surr}} \geq R_{\min} \ln 2, \quad g_k^{\text{surr}} \geq \omega_k^{\min} \ln 2, \\ & (16d), (16e). \end{aligned} \quad (22)$$

It can be solved via the BCD method by alternating between optimizing the vehicle beamformers $\{\mathbf{W}_{k,s}, \mathbf{W}_{k,c}\}$ and the relay beamformer $\mathbf{W}_r$. The procedure is summarized in **Algorithm 1**.

Algorithm 1 Proposed Beamforming Design Algorithm

- 1: **Initialize:** Iteration index $t = 0$; Initial $\{\mathbf{W}_{k,s}^{(0)}, \mathbf{W}_{k,c}^{(0)}\}$ and $\mathbf{W}_r^{(0)}$; convergence accuracy ϵ .
 - 2: **repeat**
 - 3: Update $\mathbf{E}_k^{(t+1)}$, $\mathbf{Z}_k^{(t+1)}$, $\mathbf{F}_k^{(t+1)}$ and $\mathbf{U}_k^{(t+1)}$ via (19a), (19b), (21a) and (21b) using $\mathbf{W}_{k,c}^{(t)}$ and $\mathbf{W}_r^{(t)}$;
 - 4: Update $\{\mathbf{W}_{k,s}^{(t+1)}, \mathbf{W}_{k,c}^{(t+1)}\}$ by solving problem (22);
 - 5: Update $\mathbf{W}_r^{(t+1)}$ by solving problem (22);
 - 6: $t := t + 1$;
 - 7: **until Convergence**, i.e., $|P_{\text{total}}^{(t)} - P_{\text{total}}^{(t-1)}| \leq \epsilon$.
-

4. Convergence and Complexity Analysis

Theorem 1. *Algorithm 1 converges to a stationary point of problem (16).*

Proof. The MEGLDT framework ensures that the surrogate functions tightly bound the original rate functions with identical gradients at the current operating point. Consequently, the sequential optimization of auxiliary variables and beamforming matrices guarantees a monotonic decrease in the objective value. Given that the total transmit power is strictly lower-bounded by zero, the algorithm is guaranteed to converge to a stationary point satisfying the KKT conditions of the original non-convex problem (16). ■

The computational complexity is dominated by solving the convex QCQP subproblems using interior-point methods. The complexity for the vehicle beamforming update scales as $O((KM(I+J))^{3.5})$, while the relay update scales as $O(N_r^{3.5})$. The proposed MEGLDT-BCD algorithm's main operations are the auxiliary-variable updates in (19a)–(21b), matrix multiplications/inversions, and the solution of convex QCQP subproblems in each BCD iteration. The total complexity is therefore $O(T_{\text{prop}} C_{\text{sub}})$, where C_{sub} denotes the per-iteration subproblem cost. Any matrix decomposition used by numerical linear-algebra routines for log-det evaluation is not a separate algorithmic step and does not change the dominant complexity order. WMMSE has a comparable per-iteration order, i.e., $O(T_{\text{WMMSE}} C_{\text{sub}})$; hence the practical comparison depends mainly on the number of iterations required to reach the same tolerance.

5. Simulation Results

In this section, we conduct simulations to verify the effectiveness of the proposed schemes. The main parameters are summarized in Table 1.

Table 1
Simulation parameters.

Parameter	Value
Antennas/streams	$M = N_r = N_b = 4, I = J = 2$
Number of vehicles	$K = 8$ unless otherwise stated
THz channel	$f_c = 300$ GHz, $L = 3, N_{\text{sub}} = 1$
Noise and SNR	$\sigma_r^2 = \sigma_b^2 = \sigma_v^2$, SNR = 20 dB
QoS targets	$R_{\min} = 0.2, \omega_{\min} = 0.2$ unless varied

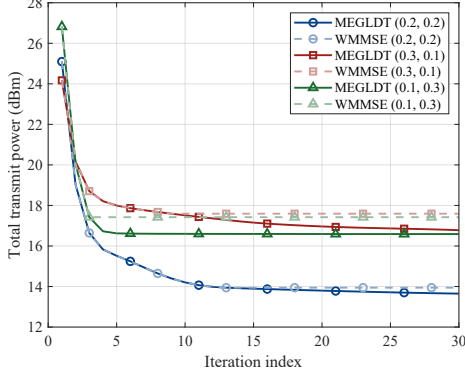


Figure 2. Convergence comparison of MEGLDT and WMMSE.

For comparison, we use WMMSE [18], ZF [19], MRT [20], and an active-RIS benchmark using a diagonal active forwarding matrix, i.e., $\Theta = \text{diag}(\phi)$, under the same active power constraint [21]. All results are averaged over 100 Monte Carlo channel realizations.

Fig. 2 compares MEGLDT-BCD and WMMSE under $(R_{\min}, \omega_{\min}) = (0.2, 0.2)$, $(0.3, 0.1)$, and $(0.1, 0.3)$. With the same feasible initialization and tolerance, MEGLDT enters the final low-power region earlier; ZF and MRT are omitted as one-shot baselines.

Fig. 3(a) illustrates the total transmit power versus the minimum communication rate $R_{\min}$. As expected, power consumption increases with $R_{\min}$ to satisfy stricter rate requirements. The proposed MEGLDT-based scheme consistently yields the lowest power consumption compared to the WMMSE, ZF, and MRT benchmarks, with the performance advantage being particularly pronounced in the lower $R_{\min}$ regime. Similarly, Fig. 3(b) depicts the impact of the minimum sensing rate $\omega_{\min}$. Increasing $\omega_{\min}$ necessitates higher power allocation. The proposed scheme again outperforms all benchmarks, offering the most significant power savings at intermediate $\omega_{\min}$ values, thereby validating its superiority in balancing sensing and communication trade-offs.

Fig. 4 examines system scalability by plotting total transmit power against the number of vehicles (K). The proposed MEGLDT algorithm consistently achieves the lowest power consumption across the entire range. For

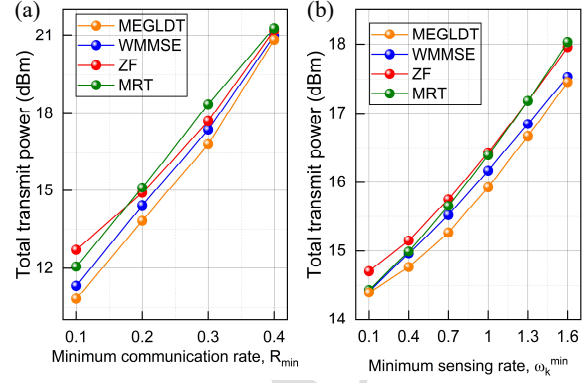


Figure 3. Total transmit power versus minimum (a) communication rate and (b) sensing rate.

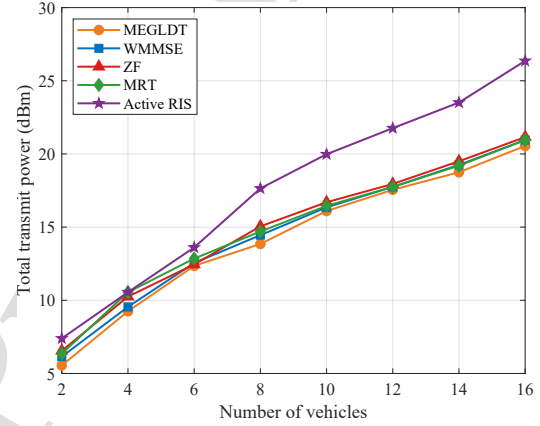


Figure 4. Total transmit power versus the number of vehicles for MEGLDT, WMMSE, ZF, MRT, and active RIS.

fairness, all schemes are evaluated under the same channel realizations and QoS thresholds, with the same active power constraint when applicable. The active-RIS benchmark requires higher power because its diagonal forwarding matrix provides less spatial processing flexibility than the full relay beamformer. The achieved communication rate and sensing metric of all plotted feasible solutions satisfy the prescribed QoS constraints within the numerical tolerance.

6. Conclusions

This letter studied relay-assisted THz V2X with integrated communication and vehicle-side sensing QoS. We formulated a joint beamforming power-minimization problem and developed an MEGLDT-BCD algorithm to handle the resulting non-convexity under communication, sensing, vehicle-power, and relay-power constraints. Simulations show power savings over WMMSE, ZF, MRT, and active RIS, with

QoS-margin checks verifying feasibility. Future work includes detection/CRB metrics, CSI uncertainty, and mobility-aware implementations.

Acknowledgements

This work was supported by the Mobile Information Networks National Science and Technology Major Project (2025ZD1303200), the Key-Area Research and Development Program of Guangdong Province (2020B010164001, 2021B0101400001), the National Key Laboratory of Information Systems Engineering (WDZC20265290412), and the National Natural Science Foundation of China (61473311).

Conflict of interest

The authors declare that there is no conflict of interest in this paper.

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CRedit Author Statement

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All authors have read and approved the final manuscript.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: